Title:

Fine-Tuning Cellpose-SAM for Bacterial Cell Segmentation in In Situ Gut Microbiota Imaging

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Date:
April 24th, 2026

Project Repository:
https://github.com/aliceh0ng/cellpose-biosensor

Model available on Hugging Face:
https://huggingface.co/aliceh0ng/cellpose-biosensor

Abstract

Accurate single-cell segmentation is a critical step in quantitative microscopy yet remains a challenge in noisy and heterogeneous environments such as in situ imaging of the gut microbiota. Generalist deep learning models, including Cellpose-SAM, have demonstrated strong cross-domain performance, but are primarily trained on eukaryotic cells and relatively clean imaging conditions, limiting their applicability to bacterial cells in complex tissue contexts. For this project, I fine-tune Cellpose-SAM for the segmentation of fluorescently labeled E. coli Nissle 1917 in murine gut tissue sections acquired using confocal microscopy. Images are characterized by high autofluorescence, uneven illumination, and small cell size, presenting significant challenges for both classical and deep learning-based segmentation approaches. Cellpose's "human-in-the-loop" annotation strategy was used to iteratively refine training data across multiple rounds, incorporating improvements in patch size and intensity normalization to enhance annotation quality and model performance. A total of 179 patches were annotated to fine-tune this model. The latest fine-tuned model achieved an average precision (AP@0.5) of 0.72 on a held-out validation image, compared to 0.62 for the base Cellpose-SAM model, with improved detection sensitivity and more accurate cell counts across diverse imaging conditions. Qualitative comparisons demonstrate enhanced segmentation in dense and low-contrast regions, although challenges related to autofluorescence persist. These results show that context specific fine-tuning of foundation segmentation models can substantially improve performance in underrepresented biological imaging contexts.

Background and Motivation

Over the past decade, advances in deep learning have reshaped image analysis pipelines in biological microscopy, enabling the automated identification and precise delineation of individual cells from complex imaging data. Accurate cell segmentation is a foundational step in quantitative microscopy, as it directly impacts downstream analyses including cell counting (e.g. quantifying bacterial abundance) and morphology characterization (e.g. assessing cancer cell shape changes). Early segmentation approaches, such as global or adaptive intensity thresholding[1], relied on image processing techniques such as edge detection[2] and watershed segmentation[3]. However, these methods often fall short in heterogenous or low signal-to-noise conditions, particularly in biological samples characterized by variable morphology or high background signal, such as autofluorescence and uneven illumination in fluorescence microscopy. As illustrated in Figure 1, applying a watershed-based segmentation pipeline to fluorescent in situ images of bacterial cells results in unsuccessful identification of bacterial cells, highlighting the limitations of intensity- and gradient-based methods in complex tissue environments. These challenges motivate the need for more robust, data-driven approaches capable of adapting to variability in both signal and morphology.

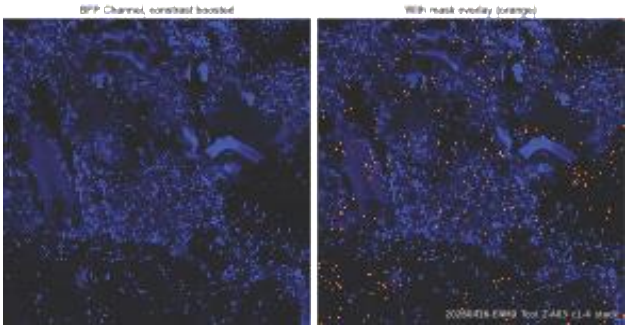


Figure 1. LNE-watershed segmentation of bacterial cells in a low SNR environment from the murine colon, contrast-boosted fluorescent image (left), overlaid with generated masks in orange (right).

The adoption of deep learning, particularly convolutional neural networks (CNNs), has significantly improved segmentation performance in more diverse contexts. A major milestone was the introduction of U-Net[4], which enabled pixel-wise classification through an encoder-decoder structure with skip connections, allowing accuracy even with limited training data. Building on this, generalist models such as Cellpose[5–7] introduced a flow-based representation of cell boundaries, improving robustness across diverse cell types without extensive parameter tuning. In parallel, methods such as StarDist[8] performs shape based detection, representing cells as star-convex polygons and achieving strong performance in densely packed or overlapping mammalian cellular environments. Released in 2025, Cellpose-SAM integrates the generalist capabilities of Cellpose with the segmentation framework of Meta's Segment Anything Model (SAM)[9], enabling zero-shot or few-shot generalization across imaging modalities. These models are trained on large and diverse image datasets and represent a significant step toward universal segmentation tools. However, despite their broad applicability, the Cellpose training data remains heavily biased toward eukaryotic cells and relatively clean imaging conditions such as pure cultures or homogenous cells.

This bias is a big limitation for applications in microbiome research and in situ imaging of the gut. Bacterial cells are substantially smaller, exhibit less distinct morphological features, and are often imaged in complex tissue environments with high background signal and autofluorescence, as shown in Figure 1. Fluorescence microscopy further introduces challenges such as uneven illumination, spectral bleed-through, and low signal-to-noise ratios. As a result, models trained predominantly on eukaryotic datasets frequently underperform in bacterial segmentation tasks, leading to missed detections and inaccurate boundary delineation. Compounding this challenge is the limited availability of high-quality, annotated datasets for bacteria in physiologically relevant contexts. Most publicly available bacterial segmentation datasets focus on in vitro conditions, such as liquid cultures or biofilms, which do not capture the spatial complexity of host-associated environments[10]. The motivation for this project is to address this gap by fine-tuning Cellpose-SAM for the segmentation of fluorescently labeled bacterial cells in in situ microscopy images of the gut microbiota. By adapting this models to a previously under-represented domain, this project aims to improve segmentation accuracy and robustness in conditions where the existing generalist models is insufficient. This is particularly relevant for quantitative studies of microbial spatial organization and activity, where accurate single-cell resolution is essential for linking bacterial behavior to microenvironmental context.

Methods and Implementation

I. Description of data

Germ-free mice were monocolonized with a genetically engineered probiotic E. coli Nissle 1917 strain. The engineered strain contains a constitutively expressed blue fluorescent protein (BFP), with two stress-induced fluorescent proteins (RFP and GFP). Upon sacrifice, whole mice colon and ileum were collected as ileum and colon rolls, with the contents preserved. Tissue was fixed in 4% PFA, followed by washes in 20% sucrose and OCT-embedding. The embedded blocks were cryo-sectioned into 8-10 um thick sections and mounted on glass slides, stained with STOX far red nuclear dye, and mounted with the ProLong glass antifade mountant. Confocal images were acquired on a Zeiss LSM 900 with Airyscan 2 in SR (super-resolution) mode using a Plan-Apochromat 100×/1.4 NA oil immersion objective. Each field of view was acquired as a 5×5 tile scan and stitched in ZEN software. Exported stacks were 16-bit unsigned integer TIFF files with four channels:

Table 1. Confocal microscopy set-up used for image acquisition

Channel	Fluorophore	Excitation/Emission	Type
C1	BFP	405 nm/410-180 nm	Constitutive reporter
C2	GFP	488 nm/490-560 nm	Stress reporter
C3	RFP	561 nm/565-640 nm	Stress reporter
C4	STOX Far Red	633 nm/650-750 nm	Nucleic acid stain

A total of 37 regions of interest were selected to be imaged across 4 mice. The physical pixel size was 0.035, and each stitched image was approximately 6,300 × 6,300 pixels (~221 × 221 μm field of view).

II. Benchmarking existing Cellpose models

To establish a baseline for fine-tuning, four pre-trained models within the Cellpose ecosystem were evaluated: Cellpose-SAM, Cellpose cyto3, bact_fluor_cp3[7], and Omnipose[11] (bact_fluor_omni). Cellpose cyto3 is a generalist model trained primarily on eukaryotic cells and serves as a standard baseline for segmentation tasks. The bact_fluor_cp3 model extends this framework with additional training on fluorescent bacterial images, improving performance on small cells under relatively controlled conditions. Omnipose modifies the underlying flow field formulation to better capture elongated and irregular morphologies, with bact_fluor_omni representing its bacteria-focused variant trained largely on in vitro datasets.

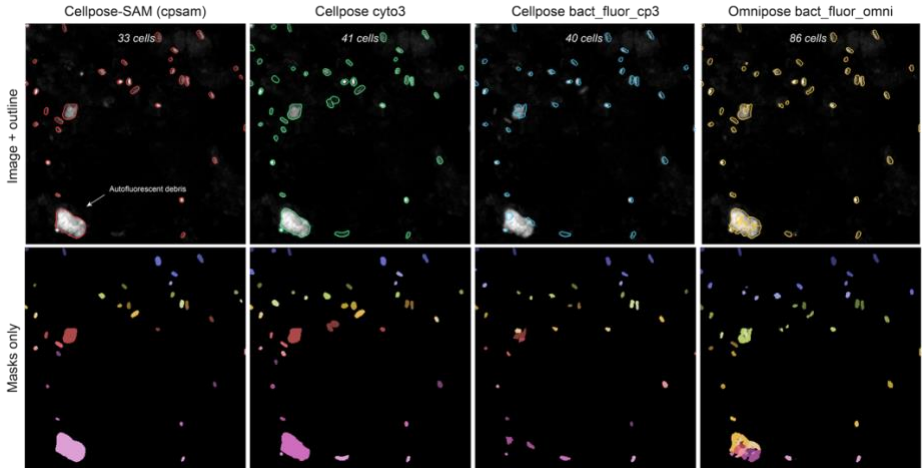


Figure 2. Benchmarking performance of Cellpose models on a 1200x1200 tile of an example confocal image.

Cellpose-SAM was selected as the base model for fine-tuning due to its broader training distribution and foundation model architecture. While bact_fluor_cp3 and bact_fluor_omni are optimized for bacterial morphology, their training data is predominantly derived from in vitro conditions and may not generalize to the high background signal and variability characteristic of in situ gut imaging. In contrast, Cellpose-SAM provides a more flexible starting point for domain adaptation, enabling improved performance on small, low signal-to-noise bacterial cells in complex tissue environments

III. Pre-processing and fine tuning methods overview

Each of the 37 image stacks were divided into a 3×3 or 5×5 grid of equal non-overlapping patches (approx. 2100 px×2100px or 1265px × 1265px). Patches at image edges absorbed any remainder pixels from integer division. Prior to segmentation, the BFP channel was normalised globally. The 1st and 99th percentiles of the full stitched BFP image were computed and used to rescale all patches to a common [0, 1] float32 range, ensuring consistent intensity scaling across patches from the same image independent of local variations in bacterial density or autofluorescence. To improve

segmentation accuracy for our specific imaging conditions, Cellpose-SAM was fine-tuned iteratively using the human-in-the-loop approach[6]. Fine-tuning was restricted to our own imaging data for this project. The base Cellpose-SAM model was pre-trained on a large and diverse corpus spanning multiple imaging modalities, so broad generalisation is already encoded in the model weights; fine-tuning on task-specific data specialises the model to our particular imaging characteristics without diluting that signal with heterogeneous external data.

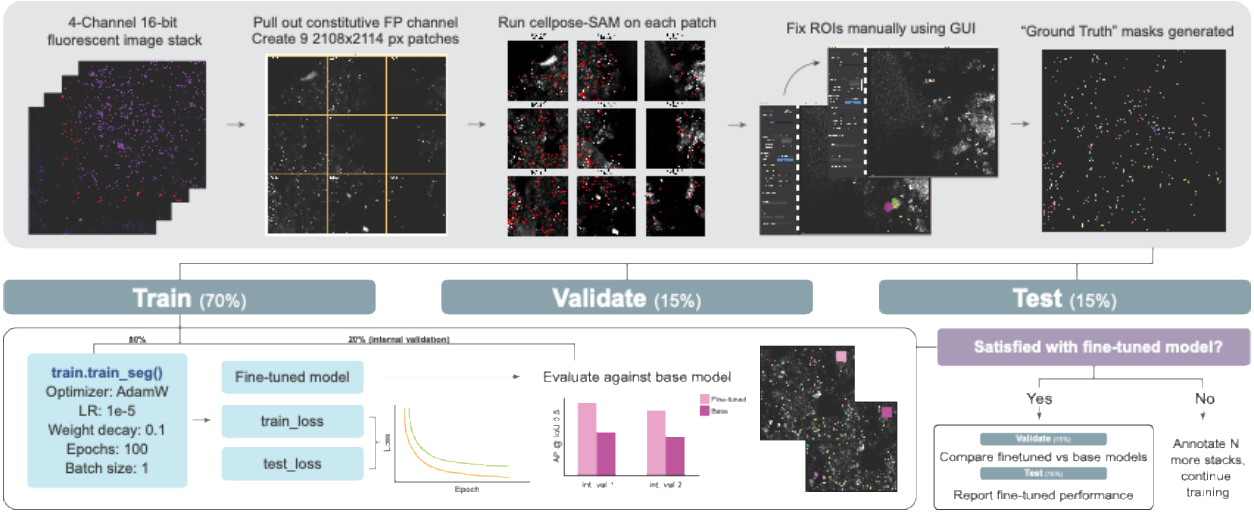


Figure 3. Overview of methodology taken to divide microscopy image into smaller patches. Top panel shows how the bacteria were annotated to set the "ground truth", followed by the split of the 37 images (and patches generated from images) into train/val/test sets. Full size versions of each image included is available in Supplemental Figures.

Annotation was performed in two rounds with refinements between rounds. In the first round ('5stacks'), training images were split into a 3×3 grid (approx 2100 × 2100 px). The base Cellpose-SAM model was run on each BFP patch to produce initial masks, which were then reviewed and corrected in the Cellpose GUI using the BFP channel alone. In the second round ('4stacks_5x5_norm'), the grid was increased to 5×5 to produce smaller patches (~1265 × 1265 px), reducing the annotation area per patch and making it easier to review individual cells carefully. Global BFP normalisation was introduced at this stage to provide a consistent intensity reference across patches. All four imaging channels (BFP, GFP, RFP, SYTOX) were exported together, and annotations were reviewed in napari to take advantage of the full spectral information: the GFP, RFP, and SYTOX channels were used to identify and exclude autofluorescent objects and host cell fragments that are difficult to distinguish from bacteria on BFP alone. Mask corrections were then applied in the Cellpose GUI, where initial masks loaded automatically alongside the multi-channel image. In both rounds, the BFP channel was the sole input to the model during training. Corrected masks from 75 patche across were used for the final fine-tuning run. Within each fine-tuning run, annotated patches were divided into an internal training set (80%) and an internal validation set (20%), stratified at the stack level (all patches from a given stack were assigned together). This internal split is independent of the stack-level validation and test splits described above. Fine-tuning used `train.train_seg` (Cellpose 4.x API) with the following hyperparameters, matching the official Cellpose-SAM training notebook.

Table 2. train.train_set() hyperparameters

Param	Value
Learning Rate	1e-5
Weight Decay	0.1
Batch size	1
Epochs	100

Segmentation inference and fine-tuning were performed on the Tropini lab's Windows computer equipped with an NVIDIA GPU (driver 560.94, CUDA 12.6) using Cellpose 4.1.1, PyTorch 2.4.0 (CUDA 12.4 build), and Python 3.10. Manual annotation was performed on a macOS workstation (Apple Silicon, MPS backend). Statistical analysis and visualisation were performed on macOS in Python 3.10.

Results and Evaluation

In total, 54 2100×2100 patches and 179 1265×1265 patches were annotated by hand using the Cellpose GUI to use for training, internal, and external validation. Upon fine-tuning and internal validation, model performance was evaluated on the internal patch-level validation set and on a held-out stack-level validation image (Scene-02) using average precision at IoU threshold 0.5 (AP@0.5). The latest fine-tuned model, 4stacks_5x5_norm (second-round annotations, 4 training stacks, 75 patches, 3,913 cells), achieved AP@0.5 = 0.724 on the validation set compared to 0.641 for the base Cellpose-SAM model. Table 2 below summarises the fine-tuning iterations to date.

Table 3. results from training and validation

	1stack (v0)	5stacks (v1)	4stacks_5x5_norm
Patch Size	2100 px×2100 px	2100 px×2100 px	1265 px×1265 px
Train Patches	7	35	75
Val Patches (internal)	2	8	25
Val Patches (external)	-	9	25
Total cells	361	4641	3913
AP@0.5 base cpsam	0.712	0.885	0.641
AP@0.5 fine-tuned cpsam	0.741	0.842	0.724
ΔAP	+0.029	-0.043	+0.083

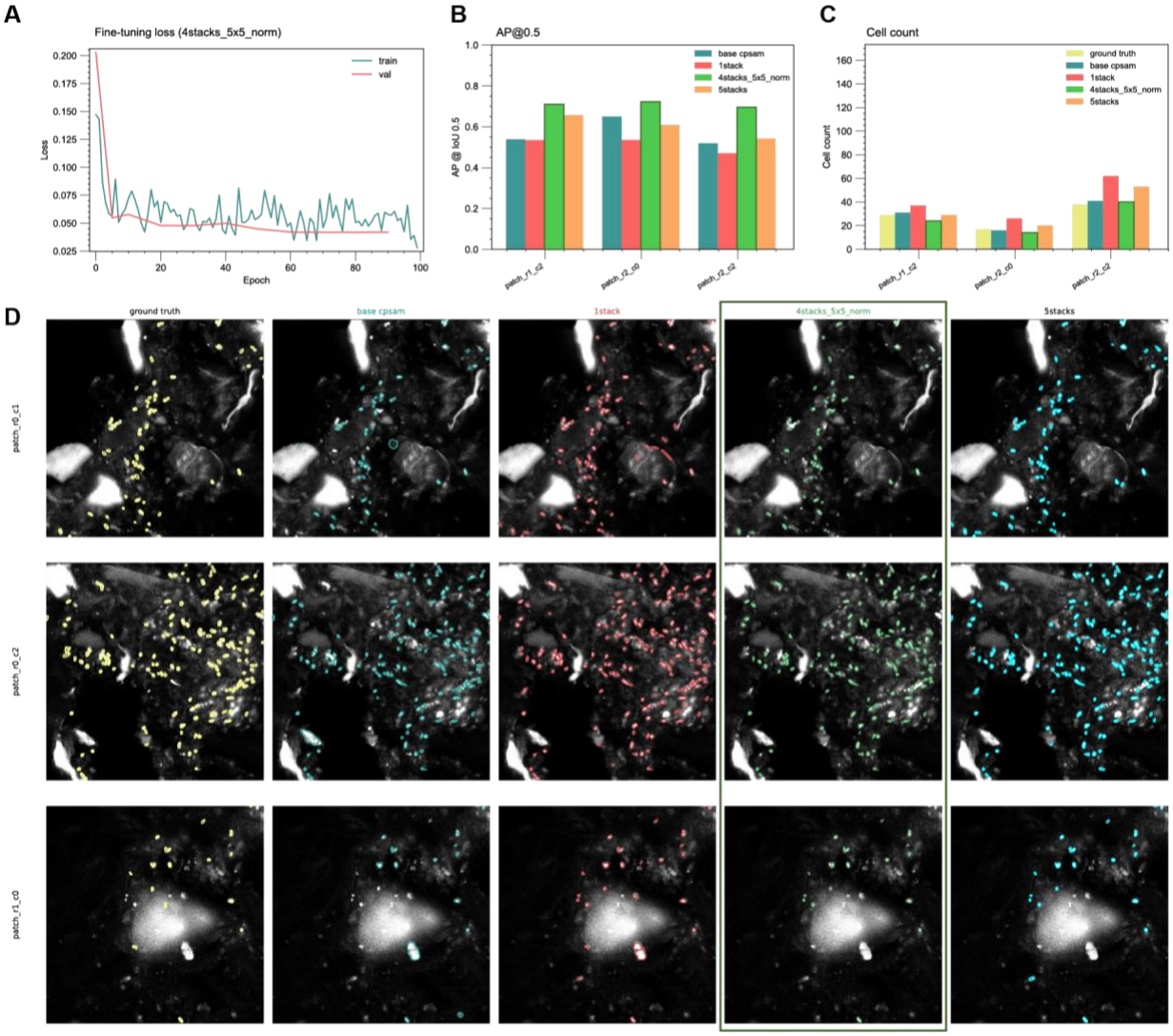


Figure 4. Results from training and initial validation of fine-tuned versions of cellpose-SAM. (A) Loss curve of 4stacks_5x5_norm, the latest fine-tuned model (B) AP@0.5 for subset of patches from Scene-02 that was used for validation across all models (C) Cell count generated by all model versions on the same patches. (D) Annotations generated by all model versions, and the "ground truth" set prior to training on the far left.

The 1stack run served as an initial feasibility test, performance metrics were recorded but should not be evaluated due to the small training set size, and the fact that patches from 1 stack were split between training and validation sets. For all future finetuning, stacks were kept together in their respective train/val/test groups. Direct comparison of ΔAP across

runs should be interpreted with caution, as each run used variable normalization and patching method as previously described. The base Cellpose-SAM AP therefore reflects the difficulty of that particular stack rather than an absolute baseline shift. Figure 4A shows the training and validation loss curves for the final model (4stacks_5x5_norm). Both losses decrease rapidly within the first ~10 epochs and stabilise thereafter, with no strong divergence between training and validation curves. This suggests that the model converges early and does not exhibit substantial overfitting despite the relatively small training dataset. Figure 4B compares segmentation performance across models using average precision at multiple IoU thresholds. The fine-tuned 4stacks_5x5_norm model consistently outperforms the base Cellpose-SAM model, with the largest improvement observed at lower IoU thresholds. The 5stacks model shows comparable to worse performance.

Discussion and Interpretation

This project demonstrates that Cellpose-SAM, despite being primarily trained on eukaryotic cell types and relatively controlled imaging conditions, can be successfully adapted to a challenging bacterial segmentation task through iterative fine-tuning. 4stacks_5x5_norm model achieving a modest but consistent improvement over base Cellpose-SAM confirms that fine-tuning on domain-specific data is beneficial even when starting from a strong generalised foundation model. The small increase in AP makes sense given the low learning rate to avoid over-fitting a foundation model already trained on tens of thousands of annotated objects. The apparent regression in the earlier 5stacks model ($\Delta AP = -0.043$) is likely due to the lower quality of the masks set as "ground truth", as larger patches were harder to annotate manually, with each stack taking 45 minutes to annotate bacteria vs. debris. The refinements introduced in the second annotation round appear to have meaningfully improved training data quality. Moving from 3×3 to 5×5 patches reduced the annotation area per session, enabling more careful review of individual cell boundaries. The introduction of global BFP normalisation provided a consistent intensity scale across patches, reducing the risk that locally normalised patches from autofluorescence-rich regions would present atypical intensity distributions to the model. Also importantly, reviewing annotations in the context of all four imaging channels allowed exclusion of autofluorescent debris that are indistinguishable from bacteria in the BFP channel alone, a source of false positive annotations that would directly degrade training signal. Together these changes suggest that annotation quality is at least as important as annotation quantity for fine-tuning performance in this setting. The cell count comparisons (Figure 4C) shows that the 4stack_5x5_norm model tends to underestimate the total number of segmented cells relative to the base model. However, this conservative bias is desirable in this context, as it reduces false positive detections arising from autofluorescent debris and background signal. As a result, the predicted counts more closely reflect true bacterial cells rather than inflated estimates driven by over-segmentation. It also is evident that the base model tends to over-segment in high-density regions and under-segment near tissue boundaries, consistent with the known challenges of detecting small, low-contrast objects against heterogeneous backgrounds. The 4stacks_5x5_norm model reduces this discrepancy in several patches, though the improvement is not uniform across all regions. This suggests that while fine-tuning improves average performance, certain patch types. Particularly those at tissue edges or with unusually high fecal autofluorescence, remain challenging and may require targeted annotation to address.

Challenges and Lessons Learned

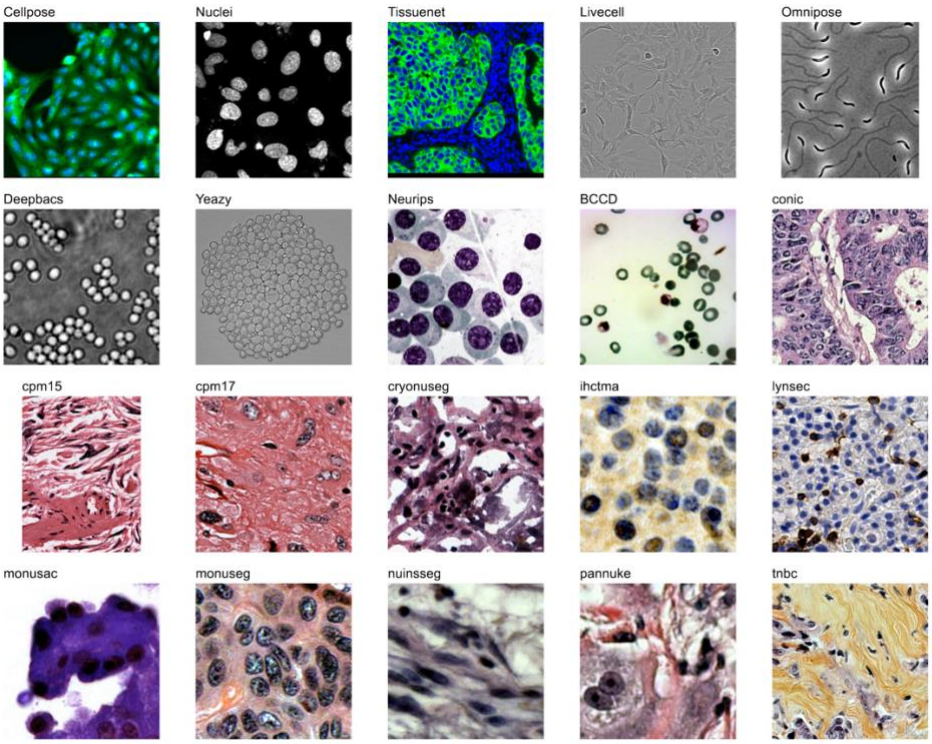
The most significant technical challenge encountered was defining which regions of interest were truly cells or background. Gut lumen content produces broad-spectrum fluorescence that is bright across the BFP, GFP, and RFP channels simultaneously, making it visually similar to bacterial cells in single-channel images. This challenge was addressed by switching to a multi-channel annotation method (on smaller patches) that allowed visual exclusion of autofluorescent objects during ground truth generation. The latter proved particularly valuable, as removing these false positives from the training set likely prevented the model from learning spurious features associated with debris morphology. A lesson learned was the importance of normalisation consistency between training and inference. In the first annotation round, per-patch normalisation meant that adjacent patches from the same image could have substantially different intensity references depending on local autofluorescence content. Switching to global normalisation in the second round aligned the intensity scale used during annotation with that used during inference, reducing a systematic mismatch that may have contributed to the 5stacks model's inconsistent validation performance. The optimal training configuration for Cellpose-SAM fine-tuning also required empirical adjustment. The SAM Vision Transformer encoder processes each image at high resolution and does not benefit from batched inference in the same way as convolutional architectures. Increasing the batch size from 1 to 4 increased per-epoch training time approximately 40-fold without a corresponding improvement in throughput, and reverting to batch_size=1 restored training speed to an acceptable level (~32 seconds per epoch on an NVIDIA GPU with 11 GB VRAM). Finally, the iterative nature of the annotation and fine-tuning workflow itself was a challenge to manage. Maintaining clear

versioning of patch datasets, model checkpoints, and validation results was essential for interpreting performance trends across runs. Establishing a structured directory layout and per-run metadata records early in the project substantially reduced ambiguity when comparing results across fine-tuning iterations.

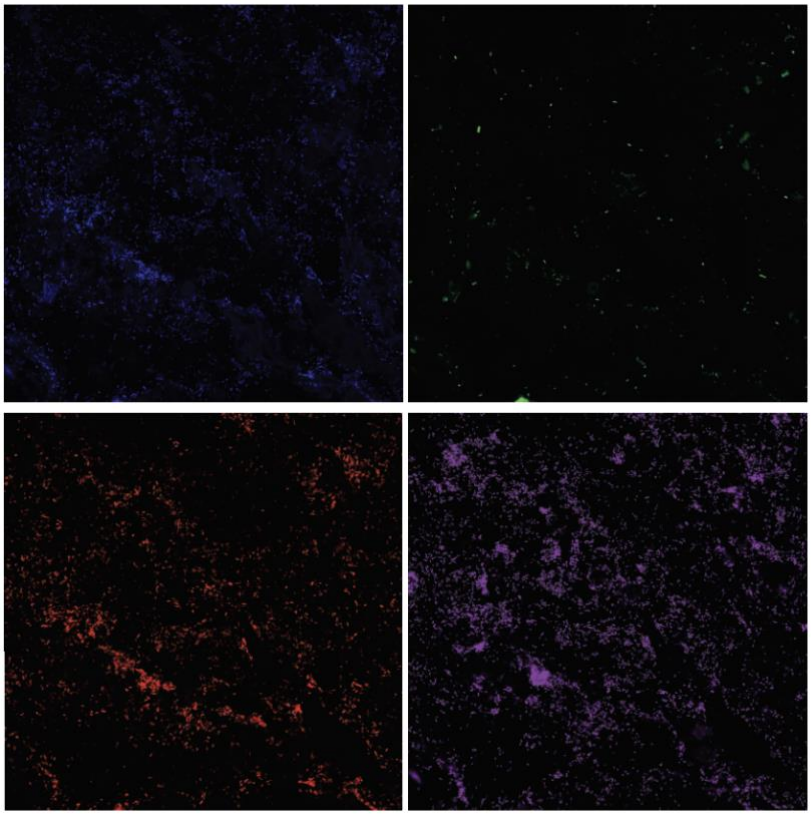
References

1. Otsu, N. A Threshold Selection Method from Gray-Level Histograms. *IEEE Trans. Syst. Man Cybern.* **9**, 62–66 (1979).
2. Accame, M. & De Natale, F. G. B. Edge detection by point classification of Canny filtered images. *Signal Process.* **60**, 11–22 (1997).
3. Beucher, S. Watershed, Hierarchical Segmentation and Waterfall Algorithm. in *Mathematical Morphology and Its Applications to Image Processing* (eds Serra, J. & Soille, P.) 69–76 (Springer Netherlands, Dordrecht, 1994). doi:10.1007/978-94-011-1040-2_10.
4. Ronneberger, O., Fischer, P. & Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (eds Navab, N., Hornegger, J., Wells, W. M. & Frangi, A. F.) 234–241 (Springer International Publishing, Cham, 2015). doi:10.1007/978-3-319-24574-4_28.
5. Stringer, C., Wang, T., Michaelos, M. & Pachitariu, M. Cellpose: a generalist algorithm for cellular segmentation. *Nat. Methods* **18**, 100–106 (2021).
6. Pachitariu, M. & Stringer, C. Cellpose 2.0: how to train your own model. *Nat. Methods* **19**, 1634–1641 (2022).
7. Stringer, C. & Pachitariu, M. Cellpose3: one-click image restoration for improved cellular segmentation. *Nat. Methods* **22**, 592–599 (2025).
8. Weigert, M. & Schmidt, U. Nuclei Instance Segmentation and Classification in Histopathology Images with Stardist. in *2022 IEEE International Symposium on Biomedical Imaging Challenges (ISBIC)* 1–4 (2022). doi:10.1109/ISBIC56247.2022.9854534.
9. Ravi, N. *et al.* SAM 2: Segment Anything in Images and Videos. Preprint at https://doi.org/10.48550/arXiv.2408.00714 (2024).
10. Ma, J. *et al.* The multimodality cell segmentation challenge: toward universal solutions. *Nat. Methods* **21**, 1103–1113 (2024).
11. Cutler, K. J. *et al.* Omnipose: a high-precision morphology-independent solution for bacterial cell segmentation. *Nat. Methods* **19**, 1438–1448 (2022).

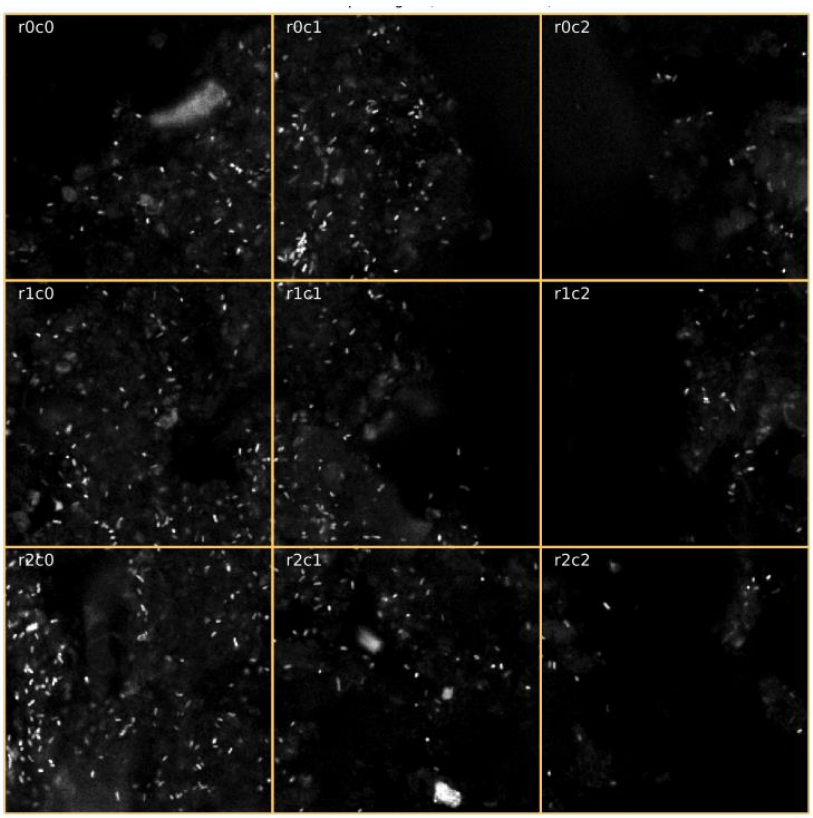
Supplemental Figures



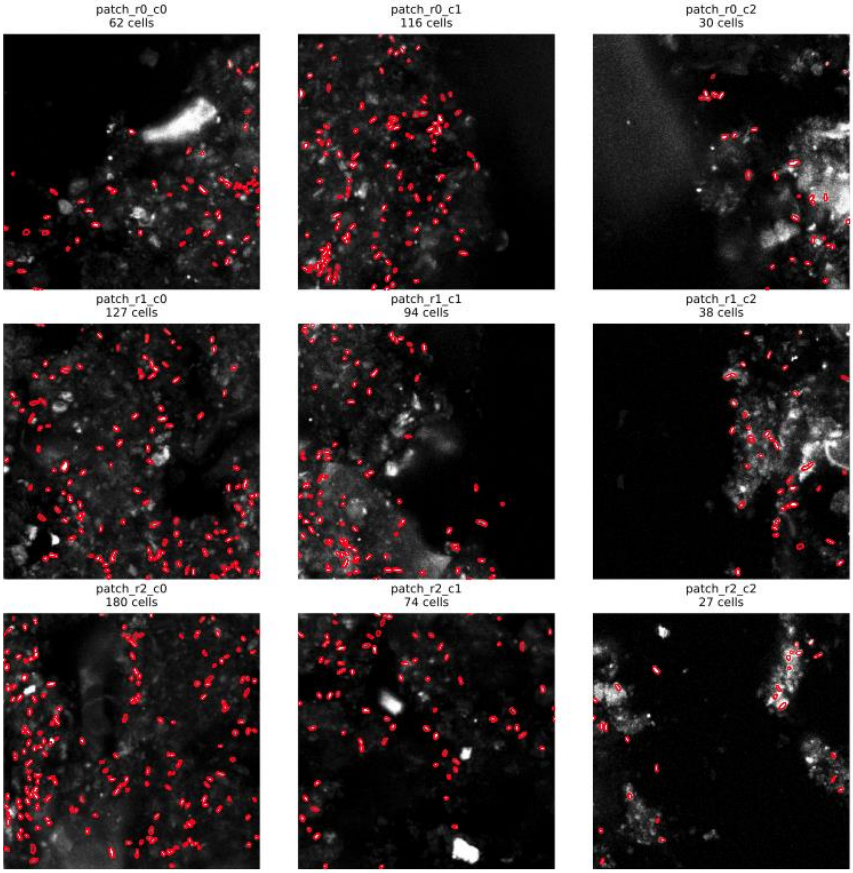
Supplemental Figure 1. Example of training images from Cellpose-SAM (Figure S1)



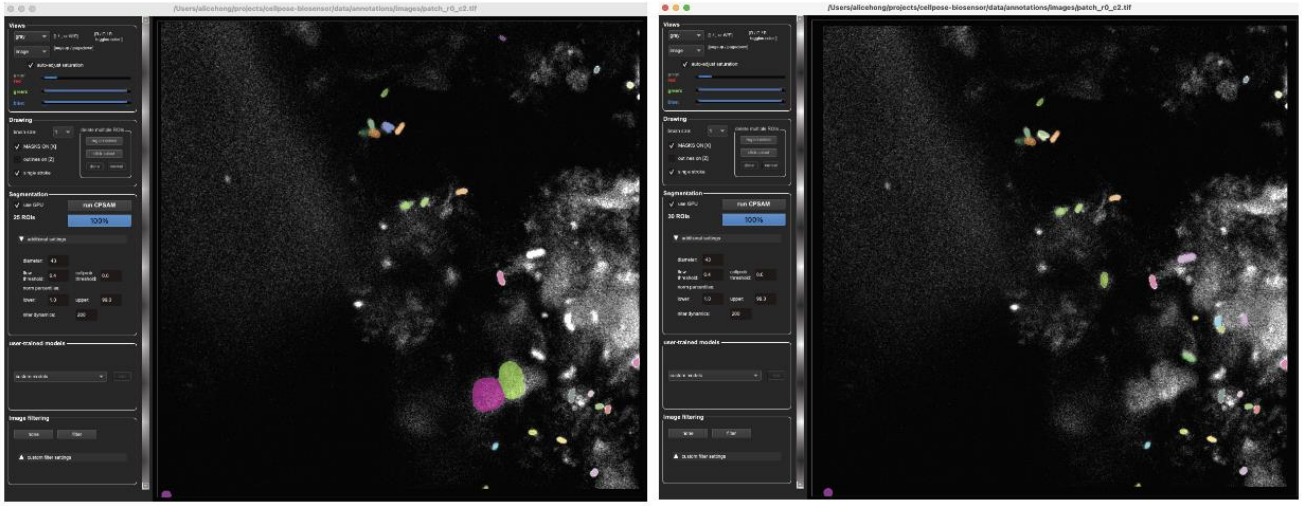
Supplemental Figure 2. Example of all 4 channels in a stack. Channel 1 BFP (on top left), channel 2 GFP (top right), channel 3 RFP (bottom left), channel 4 SYTOX (bottom right). Not contrast adjusted.



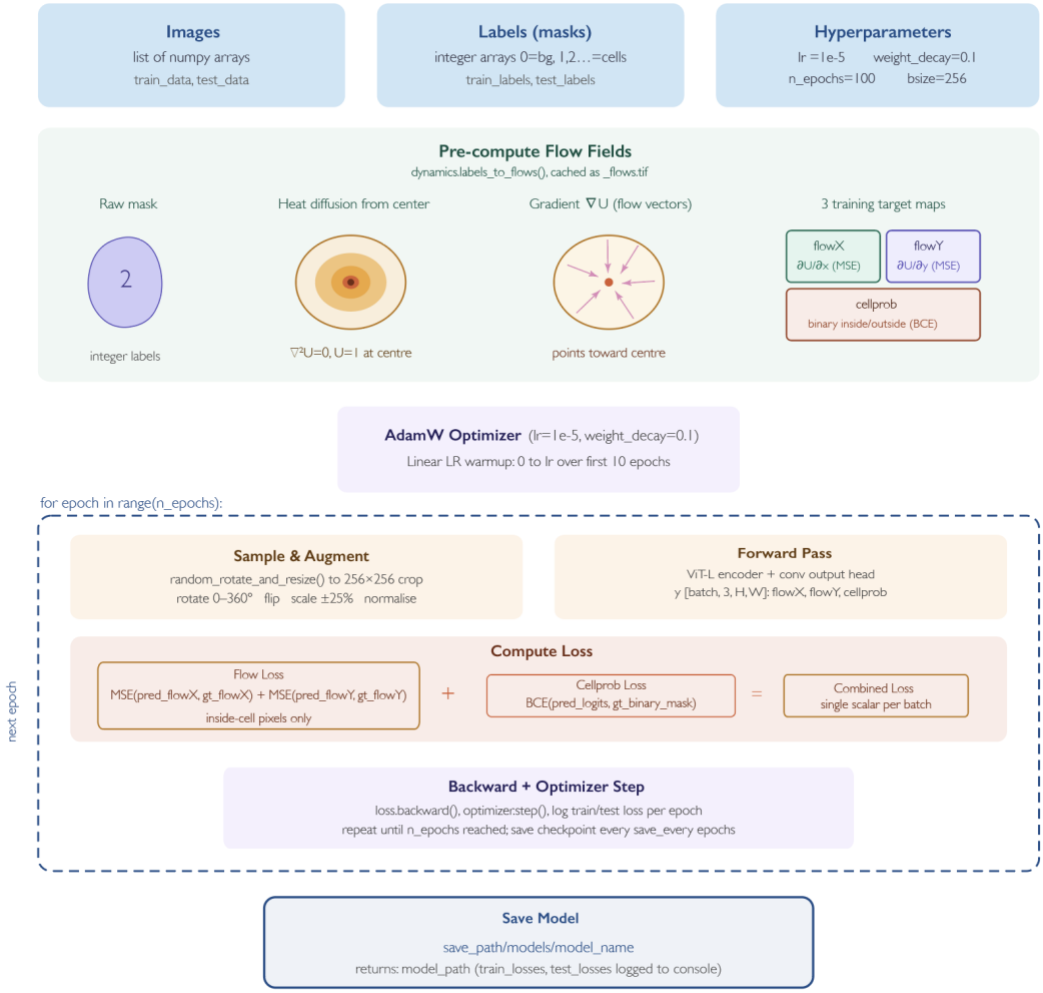
Supplemental Figure 3. Confocal image (BFP channel from S.Fig2, grayscale) split into 9 patches.



Supplemental Figure 4. 9 patches, 748 total cells segmented by base cpsam



Supplemental Figure 5. Manual annotations to create "ground truth" masks using Cellpose GUI post initial segementaiton with base callpose-SAM.



Supplemental Figure 6. Flowchart describing how train.train_seg() works to train cellpose-SAM using AdamW and low learning rate. Image created in Adobe illustrator.